Multi-Modal Chest X-Ray Intelligence System

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Course: DSAI 413
Assignment: Assignment 2 – Multi-Modal Chest X-Ray Intelligence System

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1. Architecture Overview

This project implements a dual-mode multi-modal chest X-ray intelligence system designed to perform two complementary tasks: automated report generation and RAG-based clinical question answering.

The first mode, Report Generation Mode, takes a chest X-ray image as input and uses MedGemma to generate a structured radiology report containing two sections: Findings and Impression.

The second mode, QA Mode, uses a Retrieval-Augmented Generation (RAG) pipeline. CLIP extracts visual embeddings from chest X-ray images, and these embeddings are stored in a FAISS index. When a user uploads an image and asks a clinical question, the system retrieves the most visually similar chest X-ray cases along with their generated reports. The retrieved reports are used as contextual evidence, and MedGemma produces a grounded answer based on both the image and the retrieved context.

ColPali is integrated as the mandatory multimodal retrieval model and included in the model comparison. The entire system is deployed through a Gradio application that provides two independent tabs: Report Generation and QA.

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2. Model Choices

MedGemma 4B

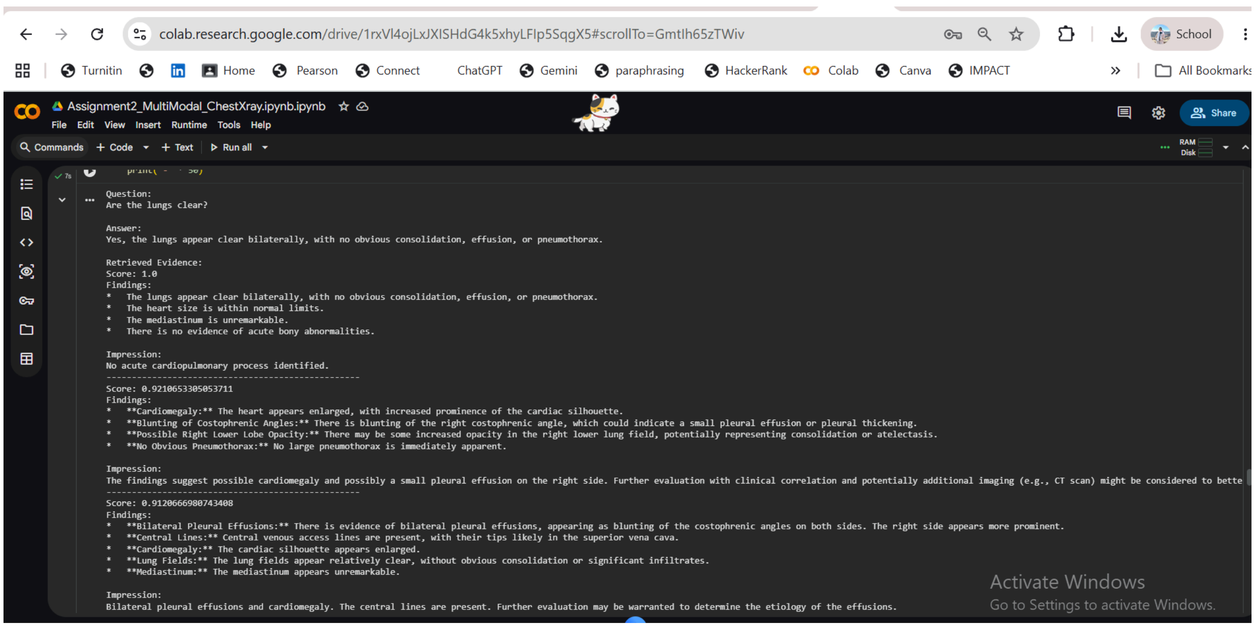
MedGemma is a medical vision-language model developed for healthcare applications. It is used as the primary reasoning model for both report generation and question answering. The model analyzes chest X-ray images and produces clinically meaningful outputs.

CLIP + FAISS

CLIP is used to extract image embeddings from chest X-ray images. These embeddings are indexed using FAISS, which enables efficient similarity search. This combination forms the retrieval component of the RAG pipeline.

ColPali

ColPali is integrated as the mandatory multimodal retrieval model required by the assignment. It is particularly strong in visual-document retrieval and is included as part of the comparative analysis.



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3. Comparison Results

MedGemma provides the strongest medical reasoning capabilities and is the most suitable model for generating structured radiology reports and answering clinical questions. Its main limitation is that it requires gated Hugging Face access and substantial GPU resources.

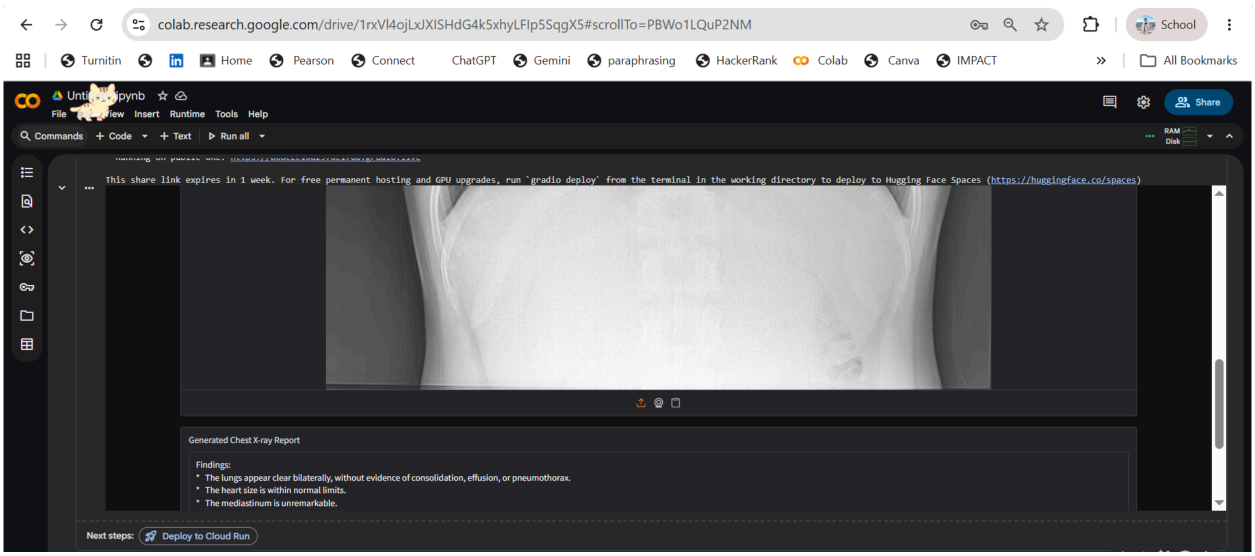
CLIP combined with FAISS is highly efficient for retrieving visually similar chest X-ray images. Although CLIP is not specifically trained for medical diagnosis, it performs well as a retrieval mechanism and significantly improves the QA pipeline.

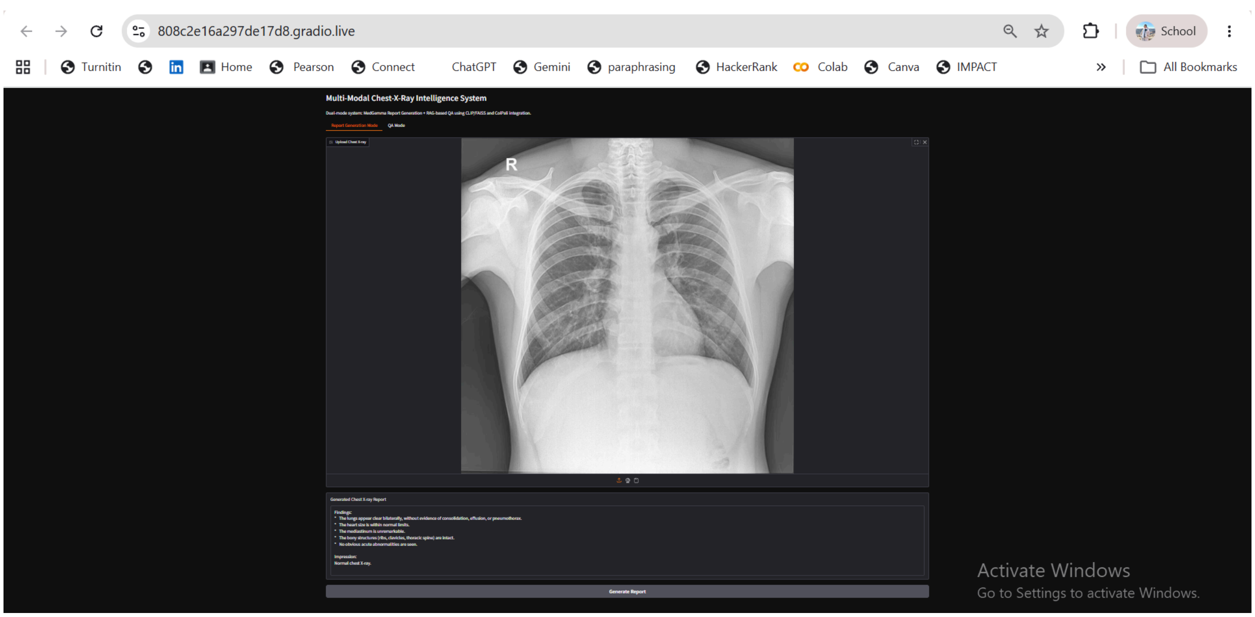
ColPali is a powerful multimodal retrieval model with strong visual-document understanding capabilities. However, it is computationally more demanding than CLIP and may require higher GPU memory to load the full model.

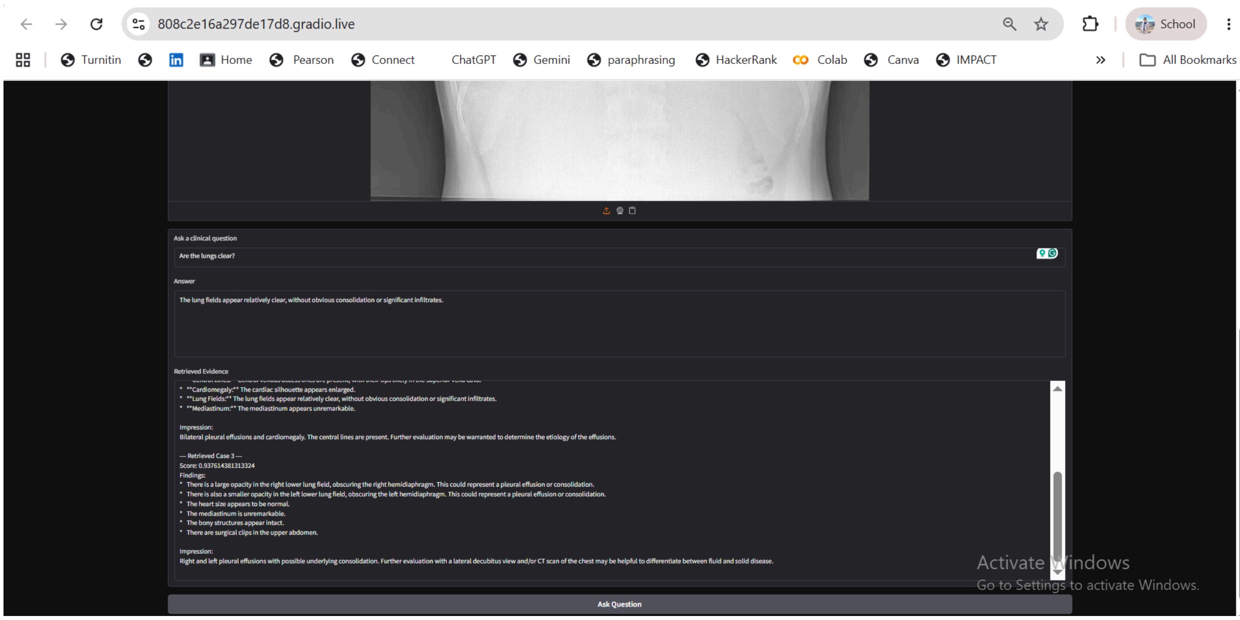
Overall, the best-performing system combines MedGemma for medical reasoning and generation with CLIP and FAISS for retrieval. ColPali is included to satisfy the assignment requirements and to provide an additional multimodal retrieval baseline.

comparison

	Model	Used For	Strength	Limitation
0	MedGemma 4B	Report Generation and QA Answering	Medical vision-language reasoning	Requires gated Hugging Face access and GPU memory
1	CLIP + FAISS	Image retrieval for RAG	Fast image similarity search	Not medically specialized
2	ColPali	Mandatory multimodal retrieval integration	Strong visual-document retrieval	Full model requires high GPU memory



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Turnitin
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Connect
ChatGPT
Gemini
paraphrasing
HackerRank
Colab
Canva
IMPACT
All Bookmarks
School
Ask a clinical question
Are the lungs clear?
Answer
The lung fields appear relatively clear, without obvious consolidation or significant infiltrates.
Retrieved Evidence
* **Cardiomegaly:** The cardiac silhouette appears enlarged.
* **Lung Fields:** The lung fields appear relatively clear, without obvious consolidation or significant infiltrates.
* **Mediastinum:** The mediastinum appears unremarkable.
Impression:
Bilateral pleural effusions and cardiomegaly. The central lines are present. Further evaluation may be warranted to determine the etiology of the effusions.
--- Retrieved Case 3 ---
Score: 0.937614381313324
Findings:
* There is a large opacity in the right lower lung field, obscuring the right hemidiaphragm. This could represent a pleural effusion or consolidation.
* There is also a smaller opacity in the left lower lung field, obscuring the left hemidiaphragm. This could represent a pleural effusion or consolidation.
* The heart size appears to be normal.
* The mediastinum is unremarkable.
* The bony structures appear intact.
* There are surgical clips in the upper abdomen.
Impression:
Right and left pleural effusions with possible underlying consolidation. Further evaluation with a lateral decubitus view and/or CT scan of the chest may be helpful to differentiate between fluid and solid disease.
Ask Question
Activate Windows
Go to Settings to activate Windows.